



Assignment for Backend Engineer Position Objective

This assignment is designed to evaluate the candidate's ability to:

1. Design and implement a clean and structured backend API
2. Understand the data flow of IoT → MQTT → Backend → Database
3. Think using event-driven and real-time system architecture
4. Write production-ready code that can be integrated into a larger system

Assignment Description

1. Sensor Data Ingestion API

The candidate is required to build a backend endpoint that receives sensor data from a greenhouse IoT system.

- **Endpoint:** POST /sensor-data
- **Payload:** Represents data from greenhouse sensors (e.g., temperature, humidity)

The backend must:

- a. Validate the incoming payload (required fields, correct data types)
- b. Store the data in a database (PostgreSQL, SQLite, or any local DB)
- c. Return a clear success or error response

2. Device Control API (MQTT Integration)

The candidate must implement an API endpoint to send control commands to greenhouse devices such as fans, vents, or pumps.

- **Endpoint:** POST /device-control
- **Payload includes:**
 - a. Device ID
 - b. Command (ON or OFF)

The backend must:

- a. Validate the request
- b. Publish the command to an MQTT broker using a structured topic
- c. Use a local MQTT broker (e.g., Mosquitto); no physical devices required

Example MQTT Topic: greenhouse/control/{device_id}

3. System Health Check Endpoint

The candidate must create a simple endpoint to report backend system status.

- **Endpoint:** GET /status
- **Response must indicate:**
 - a. Backend service status
 - b. Database connection status
 - c. MQTT connection status

4. Documentation Requirements

The candidate must include:

- README.md file explaining how to run the project
- A brief explanation covering:
 - a. The chosen system architecture
 - b. How MQTT is integrated
 - c. How errors and edge cases are handled

5. Expected Deliverables

- Complete source code
- README documentation
- A written explanation of design decisions
- A presentation file in PDF format

6. Evaluation Criteria

- Code structure and readability
- Clarity and consistency of API design
- Correct and structured MQTT usage
- Understanding of real-time system behavior
- Quality of documentation and explanation